# Rajarata University of Sri Lanka

Faculty of Applied Sciences

BSc. In Information Technology

## ICT-2204

Web Technologies & Web Design

**Mini Project Phase 01 – Project Proposal**


**Team StackTwins**

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# 1. Introduction

The Smart Study Planner is an interactive web application designed to help university students manage their academic workload efficiently. Students frequently struggle with tracking and managing multiple subjects, assignment deadlines, and exam schedules simultaneously. This application provides a centralized, visually appealing platform where students can organize their academic life, monitor their progress, and stay motivated throughout the semester.

By combining modern web technologies including HTML5, CSS3, Bootstrap 5, and JavaScript, this project aims to produce a fully functional, responsive, and aesthetically professional web application suitable for real-world use.

# 2. Purpose of the Web Application

The Smart Study Planner serves the following core purposes:

- Provide students with a digital dashboard to manage subjects, tasks, and exam dates in one place.
- Offer real-time countdown timers to keep students aware of upcoming deadlines and examinations.
- Enable task tracking with a visual progress bar so students can measure their daily study completion.
- Allow goal setting and daily habit logging for consistent academic performance.
- Present a professional, mobile-responsive interface that works across all devices.

# 3. Features to be implemented

## 3.1 Core Features (Minimum Three JavaScript Features)

| # | Feature | Description | JS Category |
|---|---|---|---|
| 1 | Exam Countdown Timer | JavaScript Date API calculates and displays live countdown to each exam date entered by user. | Dynamic Content Updates |
| 2 | Task Progress Bar | A visual progress bar updates dynamically as users mark tasks complete, showing percentage done. | Dynamic Content Updates |
| 3 | Form Validation | Add Subject and Add Task forms validate all fields (required fields, date format, character limits) before submission. | Form Validation |
| 4 | Smooth Scrolling Navigation | Navigation bar links use smooth scrolling to transition between Home, Planner, and Contact sections. | Smooth Scrolling |
| 5 | Event Handling – Modal Popups | Clicking a subject card opens a Bootstrap modal with full subject details and task list. | Event Handling |
| 6 | Custom Animations | Fade-in animations on page load for cards and sections using JavaScript class toggling with CSS transitions. | Custom Animations |

### 3.2 Additional Features

- Bootstrap 5 responsive grid – adapts layout for mobile, tablet, and desktop screens.
- Subject cards with color coding for easy visual differentiation.
- Navigation bar with brand logo, responsive hamburger menu on mobile.
- Footer with developer information and quick links.
- Bootstrap carousel on the home page showcasing productivity tips.
- Local Storage integration to persist user data between browser sessions (bonus).

## 4. Target Audience

The primary target audience for the Smart Study Planner is:

- Undergraduate and postgraduate university students who manage multiple subjects simultaneously.
- Students at Rajarata University of Sri Lanka and similar institutions who need structured academic planning tools.
- Self-directed learners who prefer digital tools for organizing their study schedules.
- Lecturers and academic advisors who may recommend productivity tools to their students.

The application is designed to be simple enough for first-year students while offering enough depth to satisfy the needs of senior undergraduates managing complex project and examination schedules.

## 5. Technology Stack

| Technology | Purpose |
|---|---|
| **HTML5** | Page structure, semantic elements, forms |
| **CSS3** | Styling, animations, transitions, custom variables |
| **Bootstrap 5** | Responsive grid, navbar, cards, modals, carousel |
| **JavaScript (Vanilla)** | Dynamic content, countdown timers, form validation, DOM manipulation |
| **Google Fonts** | Typography – Poppins display font for professional appearance |
| **GitHub Pages** | Free hosting and repository for project submission |

## 6. Page Structure

### Page 1 – Home (index.html)

- Hero section with animated welcome message and call-to-action button.
- Bootstrap carousel with study tips and motivational quotes.
- Overview cards highlighting app features (Subjects, Tasks, Countdown).
- Smooth-scroll navigation bar linking to all pages.

### Page 2 – Planner (planner.html)

- Add Subject form with name, color, and exam date fields (with validation).
- Subject cards grid with countdown timer displayed per subject.
- Task list section: add tasks, mark complete, view progress bar.
- Modal popup on card click showing full subject details.

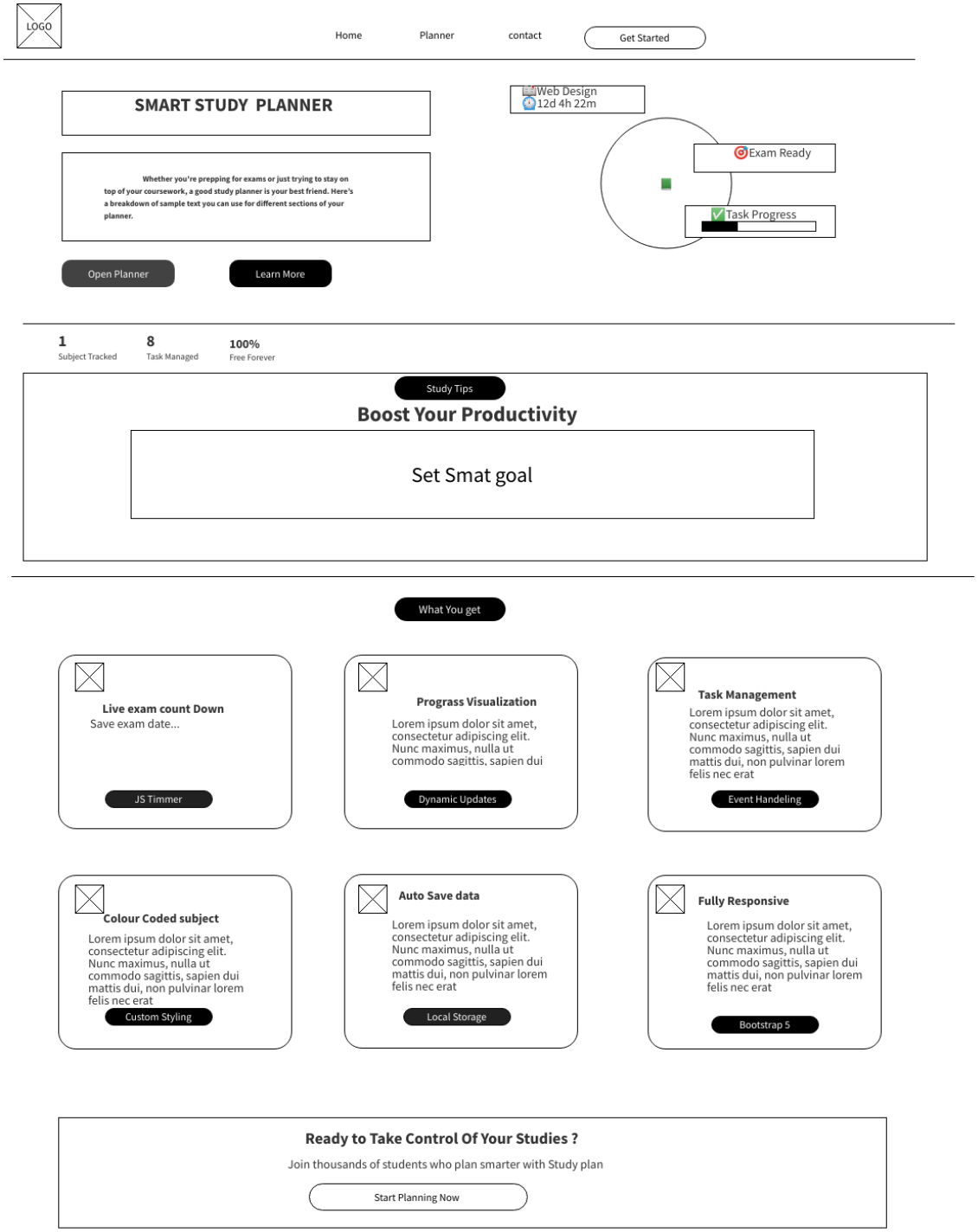
### Page 3 – Contact (contact.html)

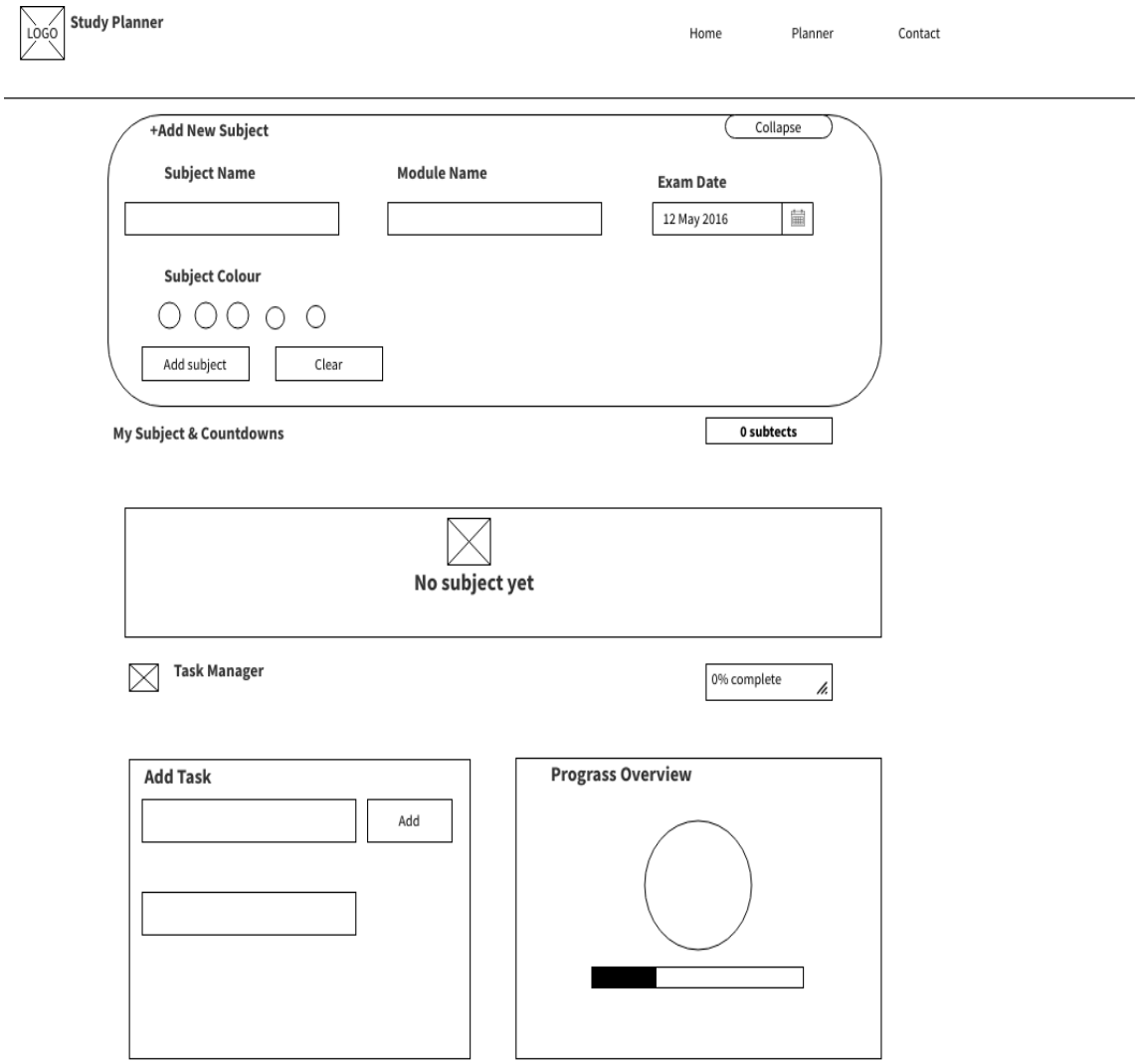
- Contact form with name, email, subject, and message fields.
- Full form validation (required fields, email format check).
- Success animation upon valid form submission.
- Developer information and links section.

## 7. Project Timeline

| Phase | Deliverable | Description |
|---|---|---|
| Phase 1 | Proposal & Wireframes | Project proposal document, wireframe sketches for all 3 pages |
| Phase 2 | HTML/CSS/Bootstrap Layout | Complete page structure, responsive design, Bootstrap components |
| Phase 3 | JavaScript & Hosting | All JS features implemented, hosted on GitHub Pages, README written |

# 8. Wireframe design




LOGO
Study Planner
Home
Planner
Contact
+Add New Subject
Collapse
Subject Name
Module Name
Exam Date
12 May 2016
Subject Colour
Add subject
Clear
My Subject & Countdowns
0 subtects
No subject yet
Task Manager
0% complete
Add Task
Add
Prograss Overview

## Get in touch

suggestions?

Send a Message

Message Sent!
Thank you for reaching out.
We'll get back to you soon.
First Name *

Your name

your last name

Email Address

Subject

Select

Message